

We replace v_2 by $3v_2$, obtaining $v_2 = 3\sqrt{x} - 2$, and compute v_3 as

$$\begin{aligned} v_3 &= x - \frac{\int_0^1 x \, dx}{\int_0^1 1 \, dx} 1 - \frac{\int_0^1 x(3\sqrt{x} - 2) \, dx}{\int_0^1 (3\sqrt{x} - 2)^2 \, dx} (3\sqrt{x} - 2) \\ &= x - \frac{1/2}{1} - \frac{6/5 - 1}{9/2 - 8 + 4} (3\sqrt{x} - 2) = x - \frac{1}{2} - \frac{2}{5} (3\sqrt{x} - 2) \\ &= x - \frac{6}{5}\sqrt{x} + \frac{3}{10}. \end{aligned}$$

Replacing v_3 by $10v_3$, we obtain the orthogonal basis

$$\{1, 3\sqrt{x} - 2, 10x - 12\sqrt{x} + 3\}.$$

SUMMARY

1. A basis for a subspace W is orthogonal if the basis vectors are mutually perpendicular, and it is orthonormal if the vectors also have length 1.
2. Any orthogonal set of vectors in $\mathbb{R}^n$ is a basis for the subspace it generates.
3. Let W be a subspace of $\mathbb{R}^n$ with an orthogonal basis. The projection of a vector b in $\mathbb{R}^n$ on W is equal to the sum of the projections of b on each basis vector.
4. Every nonzero subspace W of $\mathbb{R}^n$ has an orthonormal basis. Any basis can be transformed into an orthogonal basis by means of the Gram-Schmidt process, in which each vector a_j of the given basis is replaced by the vector v_j obtained by subtracting from a_j its projection on the subspace generated by its predecessors.
5. Any orthogonal set of vectors in a subspace W of $\mathbb{R}^n$ can be expanded, if necessary, to an orthogonal basis for W .
6. Let A be an $n \times k$ matrix of rank k . Then A can be factored as QR , where Q is an $n \times k$ matrix with orthonormal column vectors and R is a $k \times k$ upper-triangular invertible matrix.

EXERCISES

In Exercises 1–4, verify that the generating set of the given subspace W is orthogonal, and find the projection of the given vector b on W .

1. $W = \text{sp}([2, 3, 1], [-1, 1, -1]); b = [2, 1, 4]$
2. $W = \text{sp}([-1, 0, 1], [1, 1, 1]); b = [1, 2, 3]$

3. $W = \text{sp}([1, -1, -1, 1], [1, 1, 1, 1], [-1, 0, 0, 1]); b = [2, 1, 3, 1]$

4. $W = \text{sp}([1, -1, 1, 1], [-1, 1, 1, 1], [1, 1, -1, 1]); b = [1, 4, 1, 2]$

5. Find an orthonormal basis for the plane $2x + 3y + z = 0$.

6. Find an orthonormal basis for the subspace $W = \{[x_1, x_2, x_3, x_4] \mid x_1 = x_2 + 2x_3, x_4 = -x_2 + x_3\}$ of $\mathbb{R}^4$.
7. Find an orthonormal basis for the subspace $\text{sp}([0, 1, 0], [1, 1, 1])$ of $\mathbb{R}^3$.
8. Find an orthonormal basis for the subspace $\text{sp}([1, 1, 0], [-1, 2, 1])$ of $\mathbb{R}^3$.
9. Transform the basis $\{[1, 0, 1], [0, 1, 2], [2, 1, 0]\}$ for $\mathbb{R}^3$ into an orthonormal basis, using the Gram-Schmidt process.
10. Repeat Exercise 9, using the basis $\{[1, 1, 1], [1, 0, 1], [0, 1, 1]\}$ for $\mathbb{R}^3$.
11. Find an orthonormal basis for the subspace of $\mathbb{R}^4$ spanned by $[1, 0, 1, 0]$, $[1, 1, 1, 0]$, and $[1, -1, 0, 1]$.
12. Find an orthonormal basis for the subspace $\text{sp}([1, -1, 1, 0, 0], [-1, 0, 0, 0, 1], [0, 0, 1, 0, 1], [1, 0, 0, 1, 1])$ of $\mathbb{R}^5$.
13. Find the projection of $[5, -3, 4]$ on the subspace in Exercise 7, using the orthonormal basis found there.
14. Repeat Exercise 13, but use the subspace in Exercise 8.
15. Find the projection of $[2, 0, -1, 1]$ on the subspace in Exercise 11, using the orthonormal basis found there.
16. Find the projection of $[-1, 0, 0, 1, -1]$ on the subspace in Exercise 12, using the orthonormal basis found there.
17. Find an orthonormal basis for $\mathbb{R}^4$ that contains an orthonormal basis for the subspace $\text{sp}([1, 0, 1, 0], [0, 1, 1, 0])$.
18. Find an orthogonal basis for the orthogonal complement of $\text{sp}([1, -1, 3])$ in $\mathbb{R}^3$.
19. Find an orthogonal basis for the nullspace of the matrix
- $$\begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 2 & 5 & 1 & 4 \\ 1 & 1 & 2 & -1 \end{bmatrix}.$$
20. Find an orthonormal basis for $\mathbb{R}^3$ that contains the vector $(1/\sqrt{3})[1, 1, 1]$.
21. Find an orthonormal basis for $\text{sp}([2, 1, 1], [1, -1, 2])$ that contains $(1/\sqrt{6})[2, 1, 1]$.
22. Find an orthogonal basis for $\text{sp}([1, 2, 1, 2], [2, 1, 2, 0])$ that contains $[1, 2, 1, 2]$.
23. Find an orthogonal basis for $\text{sp}([2, 1, -1, 1], [1, 1, 3, 0], [1, 1, 1, 1])$ that contains $[2, 1, -1, 1]$ and $[1, 1, 3, 0]$.
24. Let B be the ordered orthonormal basis $\left(\left[\frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right], \left[\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right], \left[-\frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right] \right)$ for $\mathbb{R}^3$.
- Find the coordinate vectors $[c_1, c_2, c_3]$ for $[1, 2, -4]$ and $[d_1, d_2, d_3]$ for $[5, -3, 2]$, relative to the ordered basis B .
 - Compute $[1, 2, -4] \cdot [5, -3, 2]$, and then compute $[c_1, c_2, c_3] \cdot [d_1, d_2, d_3]$. What do you notice?
25. Mark each of the following True or False.
- All vectors in an orthogonal basis have length 1.
 - ☒ All vectors in an orthonormal basis have length 1.
 - Every nontrivial subspace of $\mathbb{R}^n$ has an orthonormal basis.
 - ☒ Every vector in $\mathbb{R}^n$ is in some orthonormal basis for $\mathbb{R}^n$.
 - Every nonzero vector in $\mathbb{R}^n$ is in some orthonormal basis for $\mathbb{R}^n$.
 - Every unit vector in $\mathbb{R}^n$ is in some orthonormal basis for $\mathbb{R}^n$.
 - Every $n \times k$ matrix A has a factorization $A = QR$, where the column vectors of Q form an orthonormal set and R is an invertible $k \times k$ matrix.
 - Every $n \times k$ matrix A of rank k has a factorization $A = QR$, where the column vectors of Q form an orthonormal set and R is an invertible $k \times k$ matrix.
 - It is advantageous to work with an orthogonal basis for W when projecting a vector $\mathbf{b}$ in $\mathbb{R}^n$ on a subspace W of $\mathbb{R}^n$.
 - It is even more advantageous to work with an orthonormal basis for W when performing the projection in part (i).

In Exercises 26–28, use the text answers for the indicated earlier exercise to find a QR-factorization of the matrix having as column vectors the transposes of the row vectors given in that exercise.

26. Exercise 7 27. Exercise 9 28. Exercise 11

29. Let A be an $n \times k$ matrix. Prove that the column vectors of A are orthonormal if and only if $A^T A = I$.
30. Let A be an $n \times n$ matrix. Prove that A has orthonormal column vectors if and only if A is invertible with inverse $A^{-1} = A^T$.
31. Let A be an $n \times n$ matrix. Prove that the column vectors of A are orthonormal if and only if the row vectors of A are orthonormal. [Hint: Use Exercise 30 and the fact that A commutes with its inverse.]

Exercises 32–35 involve inner-product spaces.

32. Let V be an inner-product space of dimension n , and let B be an ordered orthonormal basis for V . Prove that, for any vectors $\mathbf{a}$ and $\mathbf{b}$ in V , the inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ is equal to the dot product of the coordinate vectors of $\mathbf{a}$ and $\mathbf{b}$ relative to B . (See Exercise 24 for an illustration.)
33. Find an orthonormal basis for $\text{sp}(\sin x, \cos x)$ for $0 \leq x \leq \pi$ if the inner product is defined by $\langle f, g \rangle = \int_0^\pi f(x)g(x) dx$.
34. Find an orthonormal basis for $\text{sp}(1, x, x^2)$ for $-1 \leq x \leq 1$ if the inner product is defined by $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$.

35. Find an orthonormal basis for $\text{sp}(1, e^x)$ for $0 \leq x \leq 1$ if the inner product is defined by $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$.

The routine **QRFACTOR** in **LINTEK** allows the user to enter k independent row vectors in $\mathbb{R}^n$ for n and k at most 10. The program can then be used to find an orthonormal set of vectors spanning the same subspace. It will also exhibit a QR-factorization of the $n \times k$ matrix A having the entered vectors as column vectors.

For an $n \times k$ matrix A of rank k , the command $[Q, R] = \text{qr}(A)$ in **MATLAB** produces an $n \times n$ matrix Q whose columns form an orthonormal basis for $\mathbb{R}^n$ and an $n \times k$ upper-triangular matrix R (that is, with $r_{ij} = 0$ for $i > j$) such that $A = QR$. The first k columns of Q comprise the $n \times k$ matrix Q described in Corollary 1 of Theorem 6.4, and R is the $k \times k$ matrix R described in Corollary 1 with $n - k$ rows of zeros supplied at the bottom to make it the same size as A .

In Exercises 36–38, use **MATLAB** or **LINTEK** as just described to check the answers you gave for the indicated preceding exercise. (Note that the order you took for the vectors in the Gram–Schmidt process in those exercises must be the same as the order in which you supply them in the software to be able to check your answers.)

36. Exercises 7–12 37. Exercises 17, 20–23
38. Exercises 26–28

6.3

ORTHOGONAL MATRICES

Let A be the $n \times n$ matrix with column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$. Recall that these vectors form an orthonormal basis for $\mathbb{R}^n$ if and only if

$$\mathbf{a}_i \cdot \mathbf{a}_j = \begin{cases} 0 & \text{if } i \neq j, \mathbf{a}_i \perp \mathbf{a}_j \\ 1 & \text{if } i = j, \|\mathbf{a}_i\| = 1 \end{cases}$$

Because

$$A^T A = \begin{bmatrix} \text{---} \mathbf{a}_1 \text{---} \\ \text{---} \mathbf{a}_2 \text{---} \\ \vdots \\ \text{---} \mathbf{a}_n \text{---} \end{bmatrix} \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{bmatrix}$$

has $\mathbf{a}_i \cdot \mathbf{a}_j$ in the i th row and j th column, we see that the columns of A form an orthonormal basis of $\mathbb{R}^n$ if and only if

$$A^T A = I. \quad (1)$$

In computations with matrices using a computer, it is desirable to use matrices satisfying Eq. (1) as much as possible, as we discuss later in this section.

DEFINITION 6.4 Orthogonal Matrix

An $n \times n$ matrix A is **orthogonal** if $A^T A = I$.

The term *orthogonal* applied to a matrix is just a bit misleading. For example, $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ is not an orthogonal matrix. For an orthogonal matrix, not only must the columns be mutually orthogonal, they must also be unit vectors; that is, they must have length 1. This is not indicated by the name. It is unfortunate that the conventional name is *orthogonal matrix* rather than *orthonormal matrix*, but it is very difficult to change established terminology.

From Definition 6.4 we see that an $n \times n$ matrix A is orthogonal if and only if A is invertible and $A^{-1} = A^T$. Because every invertible matrix commutes with its inverse, it follows that $AA^T = I$, too; that is, $(A^T)^T A^T = I$. This means that the column vectors of A^T , which are the row vectors of A , also form an orthonormal basis for $\mathbb{R}^n$. Conversely, if the row vectors of an $n \times n$ matrix A form an orthonormal basis for $\mathbb{R}^n$, then so do the column vectors. We summarize these remarks in a theorem.

THEOREM 6.5 Characterizing Properties of an Orthogonal Matrix

Let A be an $n \times n$ matrix. The following conditions are equivalent:

1. The rows of A form an orthonormal basis for $\mathbb{R}^n$.
2. The columns of A form an orthonormal basis for $\mathbb{R}^n$.
3. The matrix A is orthogonal—that is, invertible with $A^{-1} = A^T$.

EXAMPLE 1 Verify that the matrix

$$A = \frac{1}{7} \begin{bmatrix} 2 & 3 & 6 \\ 3 & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix}$$

is an orthogonal matrix, and find A^{-1} .

SOLUTION We have

$$A^T A = \frac{1}{49} \begin{bmatrix} 2 & 3 & 6 \\ 3 & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix} \begin{bmatrix} 2 & 3 & 6 \\ 3 & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

In this example, A is symmetric, so

$$A^{-1} = A^T = A = \frac{1}{7} \begin{bmatrix} 2 & 3 & 6 \\ 3 & -6 & 2 \\ 6 & 2 & -3 \end{bmatrix}.$$

We now present the properties of orthogonal matrices that make them especially desirable for use in matrix computations.

THEOREM 6.6 Properties of Ax for an Orthogonal Matrix A

Let A be an orthogonal $n \times n$ matrix and let $\mathbf{x}$ and $\mathbf{y}$ be any column vectors in $\mathbb{R}^n$.

- | | |
|---|-----------------------------|
| 1. $(A\mathbf{x}) \cdot (A\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$. | Preservation of dot product |
| 2. $\ A\mathbf{x}\ = \ \mathbf{x}\ $. | Preservation of length |
| 3. The angle between nonzero vectors $\mathbf{x}$ and $\mathbf{y}$ equals the angle between $A\mathbf{x}$ and $A\mathbf{y}$. | Preservation of angle |

PROOF For property 1, we need only recall that the dot product $\mathbf{x} \cdot \mathbf{y}$ of two column vectors can be found by using the matrix multiplication $(\mathbf{x}^T)\mathbf{y}$. Because A is orthogonal, we know that $A^T A = I$, so

$$[(A\mathbf{x}) \cdot (A\mathbf{y})] = (A\mathbf{x})^T A\mathbf{y} = \mathbf{x}^T A^T A\mathbf{y} = \mathbf{x}^T I\mathbf{y} = \mathbf{x}^T \mathbf{y} = [\mathbf{x} \cdot \mathbf{y}].$$

HISTORICAL NOTE PROPERTIES OF ORTHOGONAL MATRICES for square systems of coefficients appear in various works of the early nineteenth century. For example, in 1833 Carl Gustav Jacob Jacobi (1804–1851) sought to find a linear substitution

$$y_1 = \sum \alpha_{1i} x_i, \quad y_2 = \sum \alpha_{2i} x_i, \dots, \quad y_n = \sum \alpha_{ni} x_i$$

such that $\sum y_i^2 = \sum x_i^2$. He found that the coefficients of the substitution must satisfy the orthogonality property

$$\sum_i \alpha_{ij} \alpha_{ik} = \begin{cases} 0, & j \neq k, \\ 1, & j = k. \end{cases}$$

One can even trace orthogonal systems of coefficients back to seventeenth- and eighteenth-century works in analytic geometry, when rotations of the plane or of 3-space are given in order to transform the equations of curves or surfaces. Expressed as matrices, these rotations would give orthogonal ones.

The formal definition of an orthogonal matrix, however, and a comprehensive discussion appeared in an 1878 paper of Georg Ferdinand Frobenius (1849–1917) entitled "On Linear Substitutions and Bilinear Forms." In particular, Frobenius dealt with the eigenvalues of such a matrix.

Frobenius, who was a full professor in Zurich and later in Berlin, made his major mathematical contribution in the area of group theory. He was instrumental in developing the concept of an abstract group, as well as in investigating the theory of finite matrix groups and group characters.

For property 2, the length of a vector can be defined in terms of the dot product—namely, $\|x\| = \sqrt{x \cdot x}$. Because multiplication by A preserves the dot product, it must preserve the length.

For property 3, the angle between nonzero vectors x and y can be defined in terms of the dot product—namely, as

$$\cos^{-1}\left(\frac{x \cdot y}{\sqrt{x \cdot x} \sqrt{y \cdot y}}\right).$$

Because multiplication by A preserves the dot product, it must preserve angles. $\blacktriangle$

EXAMPLE 2 Let v be a vector in $\mathbb{R}^3$ with coordinate vector $[2, 3, 5]$ relative to some ordered orthonormal basis (a_1, a_2, a_3) of $\mathbb{R}^3$. Find $\|v\|$.

SOLUTION We have $v = 2a_1 + 3a_2 + 5a_3$, which can be expressed as

$$v = Ax, \quad \text{where } A = \begin{bmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

Using property (ii) of Theorem 6.6, we obtain

$$\|v\| = \|Ax\| = \|x\| = \sqrt{4 + 9 + 25} = \sqrt{38}.$$

Property 2 of Theorem 6.6 is the reason that it is desirable to use orthogonal matrices in matrix computations on a computer. Suppose, for example, that we have occasion to perform a multiplication Ax for a square matrix A and a column vector x whose components are quantities we have to measure. Our measurements are apt to have some error, so rather than using the true vector x for these measured quantities, we probably work with $x + e$, where e is a nonzero error vector. Upon multiplication by A , we then obtain

$$A(x + e) = Ax + Ae.$$

The new error vector is Ae . If the matrix A is orthogonal, we know that $\|Ae\| = \|e\|$, so the magnitude of the error vector remains the same under multiplication by A . We express this important fact as follows:

Multiplication by orthogonal matrices is a *stable* operation.

If A is not orthogonal, $\|Ae\|$ can be a great deal larger than $\|e\|$. Repeated multiplication by nonorthogonal matrices can cause the error vector to blow up to such an extent that the final answer is meaningless. To take advantage

of this stability, scientists try to *orthogonalize* computational algorithms with matrices, in order to produce reliable results.

Orthogonal Diagonalization of Real Symmetric Matrices

Recall that a symmetric matrix is an $n \times n$ square matrix A in which the k th row vector is equal to the k th column vector for each $k = 1, 2, \dots, n$. Equivalently, A is symmetric if and only if it is equal to its transpose A^T . The problem of diagonalizing a real symmetric matrix arises in many applications. (See Section 8.1, for example.) As we stated in Theorem 5.5, this diagonalization can always be achieved. That is, there is an invertible matrix C such that $C^{-1}AC = D$ is a diagonal matrix. In fact, C can be chosen to be an *orthogonal matrix*, as we will show. This means that diagonalization of real symmetric matrices is a computationally stable process. We begin by proving the perpendicularity of eigenvectors of a real symmetric matrix that correspond to distinct eigenvalues.

THEOREM 6.7 Orthogonality of Eigenspaces of a Real Symmetric Matrix

Eigenvectors of a real symmetric matrix that correspond to different eigenvalues are orthogonal. That is, the eigenspaces of a real symmetric matrix are orthogonal.

PROOF Let A be an $n \times n$ symmetric matrix, and let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors corresponding to distinct eigenvalues λ_1 and λ_2 , respectively. Writing vectors as column vectors, we have

$$A\mathbf{v}_1 = \lambda_1\mathbf{v}_1 \quad \text{and} \quad A\mathbf{v}_2 = \lambda_2\mathbf{v}_2.$$

We want to show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. We begin by showing that $\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2) = \lambda_2(\mathbf{v}_1 \cdot \mathbf{v}_2)$. We compute

$$\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2) = (\lambda_1\mathbf{v}_1) \cdot \mathbf{v}_2 = (A\mathbf{v}_1) \cdot \mathbf{v}_2.$$

The final dot product can be written in matrix form as

$$(A\mathbf{v}_1)^T \mathbf{v}_2 = (\mathbf{v}_1^T A^T) \mathbf{v}_2.$$

Therefore,

$$[\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2)] = \mathbf{v}_1^T A^T \mathbf{v}_2. \quad (2)$$

On the other hand,

$$[\lambda_2(\mathbf{v}_1 \cdot \mathbf{v}_2)] = [\mathbf{v}_1 \cdot (\lambda_2\mathbf{v}_2)] = \mathbf{v}_1^T A \mathbf{v}_2. \quad (3)$$

Because $A = A^T$, Eqs. (2) and (3) show that

$$\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2) = \lambda_2(\mathbf{v}_1 \cdot \mathbf{v}_2) \quad \text{or} \quad (\lambda_1 - \lambda_2)(\mathbf{v}_1 \cdot \mathbf{v}_2) = 0.$$

Because $\lambda_1 - \lambda_2 \neq 0$, we conclude that $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, which shows that $\mathbf{v}_1$ is orthogonal to $\mathbf{v}_2$. $\blacktriangle$

The results stated for real symmetric matrices in Section 5.2 tell us that an $n \times n$ real symmetric matrix A has only real numbers as the roots of its characteristic polynomial, and that the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity; therefore, we can find a basis for $\mathbb{R}^n$ consisting of eigenvectors of A . Using the Gram-Schmidt process, we can modify the vectors of the basis in each *eigenspace* to be an orthonormal set. Theorem 6.7 then tells us that the basis vectors from different eigenspaces are also perpendicular, so we obtain a basis of mutually perpendicular real eigenvectors of unit length. We can take as the diagonalizing matrix C , such that $C^{-1}AC = D$, an orthogonal matrix whose column vectors consist of the vectors in this orthonormal basis for $\mathbb{R}^n$. We summarize our discussion in a theorem.

THEOREM 6.8 Fundamental Theorem of Real Symmetric Matrices

Every real symmetric matrix A is diagonalizable. The diagonalization $C^{-1}AC = D$ can be achieved by using a real orthogonal matrix C .

The converse of Theorem 6.8 is also true. If $D = C^{-1}AC$ is a diagonal matrix and C is an orthogonal matrix, then A is symmetric. (See Exercise 24.) The equation $D = C^{-1}AC$ is said to be an **orthogonal diagonalization** of A .

EXAMPLE 3 Find an orthogonal diagonalization of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

SOLUTION The eigenvalues of A are the roots of the characteristic equation

$$\begin{vmatrix} 1 - \lambda & 2 \\ 2 & 4 - \lambda \end{vmatrix} = \lambda^2 - 5\lambda = 0.$$

They are $\lambda_1 = 0$ and $\lambda_2 = 5$. We proceed to find the corresponding eigenvectors.

For $\lambda_1 = 0$, we have

$$A - \lambda_1 I = A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix},$$

which yields the eigenspace

$$E_0 = \text{sp}\left(\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right).$$

For $\lambda_2 = 5$, we have

$$A - \lambda_2 I = A - 5I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix},$$

which yields the eigenspace

$$E_5 = \text{sp}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right).$$

Thus,

$$\begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

is a diagonalization of A , and

$$\begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

is an orthogonal diagonalization of A . ■

EXAMPLE 4 Find an orthogonal diagonalization of the matrix

$$A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix},$$

that is, find an orthogonal matrix C such that $C^{-1}AC$ is a diagonal matrix D .

SOLUTION Example 3 of Section 5.3 shows that the eigenvalues and associated eigenspaces of A are

$$\lambda_1 = -1, \quad E_1 = \text{sp}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right),$$

$\mathbf{v}_1$

$$\lambda_2 = \lambda_3 = 2, \quad E_2 = \text{sp}\left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right).$$

$\mathbf{v}_2 \quad \mathbf{v}_3$

Notice that vectors $\mathbf{v}_2$ and $\mathbf{v}_3$ in E_2 are orthogonal to the vector $\mathbf{v}_1$ in E_1 , as must be the case for this symmetric matrix A . The vectors $\mathbf{v}_2$ and $\mathbf{v}_3$ in E_2 are not orthogonal, but we can use the Gram-Schmidt process to find an orthogonal basis for E_2 . We replace $\mathbf{v}_3$ by

$$\mathbf{v}_3 - \frac{\mathbf{v}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix}, \quad \text{or by } \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}.$$

Thus $\{[1, 1, 1], [-1, 1, 0], [-1, -1, 2]\}$ is an orthogonal basis for $\mathbb{R}^3$ of eigenvectors of A . An orthogonal diagonalizing matrix C is obtained by normalizing these vectors and taking the vectors in the resulting orthonormal basis as column vectors in C . We obtain

$$C = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}. \quad \blacksquare$$

Orthogonal Linear Transformations

Every theorem about matrices has an interpretation for linear transformations. Let A be an orthogonal $n \times n$ matrix, and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(\mathbf{x}) = A\mathbf{x}$. Theorem 6.6 immediately establishes the following properties of T :

1. $T(\mathbf{x}) \cdot T(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$; Preservation of dot product
2. $\|T(\mathbf{x})\| = \|\mathbf{x}\|$; Preservation of length
3. The angle between $T(\mathbf{x})$ and $T(\mathbf{y})$ equals the angle between $\mathbf{x}$ and $\mathbf{y}$. Preservation of angle

The first property is commonly used to define an orthogonal linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$.

DEFINITION 6.5 Orthogonal Linear Transformation

A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is **orthogonal** if it satisfies $T(\mathbf{v}) \cdot T(\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$ for all vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$.

For example, the linear transformation that reflects the plane $\mathbb{R}^2$ in a line containing the origin clearly preserves both the angle θ between vectors $\mathbf{u}$ and $\mathbf{v}$ and the magnitudes of the vectors. Because the dot product in $\mathbb{R}^2$ satisfies

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta,$$

it follows that dot products are also preserved. Therefore, this reflection of the plane is an orthogonal linear transformation.

We just showed that every orthogonal matrix gives rise to an orthogonal linear transformation of $\mathbb{R}^n$ into itself. The converse is also true.

THEOREM 6.9 Orthogonal Transformations vis-à-vis Matrices

A linear transformation T of $\mathbb{R}^n$ into itself is orthogonal if and only if its standard matrix representation A is an orthogonal matrix.

PROOF It remains for us to show that A is an orthogonal matrix if T preserves the dot product. The columns of A are the vectors $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, $\dots$, $T(\mathbf{e}_n)$, where $\mathbf{e}_j$ is the j th unit coordinate vector of $\mathbb{R}^n$. We have

$$T(\mathbf{e}_i) \cdot T(\mathbf{e}_j) = \mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j, \end{cases}$$

showing that the columns of A form an orthonormal basis of $\mathbb{R}^n$. Thus, A is an orthogonal matrix. $\blacktriangle$

EXAMPLE 5 Show that the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x_1, x_2, x_3]) = [x_1/\sqrt{2} + x_3/\sqrt{2}, x_2, -x_1/\sqrt{2} + x_3/\sqrt{2}]$ is orthogonal.

SOLUTION The orthogonality of the transformation follows from the fact that the standard matrix representation

$$A = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

is an orthogonal matrix. ■

In Exercise 40, we ask you to show that a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is orthogonal if and only if T maps unit vectors into unit vectors. Sometimes, this is an easy condition to verify, as our next example illustrates.

EXAMPLE 6 Show that the linear transformation that rotates the plane counterclockwise through any angle is an orthogonal linear transformation.

SOLUTION A rotation of the plane preserves the lengths of all vectors—in particular, unit vectors. Example 1 in Section 2.4 shows that a rotation is a linear transformation, and Exercise 40 then shows that the transformation is orthogonal. ■

You might conjecture that a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is orthogonal if and only if it preserves the angle between vectors. Exercise 38 asks you to give a counterexample, showing that this is *not* the case.

For students who studied inner-product spaces in Section 3.5, we mention that a linear transformation $T: V \rightarrow V$ of an inner-product space V is defined to be **orthogonal** if $\langle T(\mathbf{v}), T(\mathbf{w}) \rangle = \langle \mathbf{v}, \mathbf{w} \rangle$ for all vectors $\mathbf{v}, \mathbf{w} \in V$. This is the natural extension of Definition 6.5.

SUMMARY

1. A square $n \times n$ matrix A is orthogonal if it satisfies any one (and hence all) of these three equivalent conditions:
 - a. The rows of A form an orthonormal basis for $\mathbb{R}^n$.
 - b. The columns of A form an orthonormal basis for $\mathbb{R}^n$.
 - c. The matrix A is invertible, and $A^{-1} = A^T$.
2. Multiplication of column vectors in $\mathbb{R}^n$ on the left by an $n \times n$ orthogonal matrix preserves length, dot product, and the angle between vectors. Such multiplication is computationally stable.
3. A linear transformation of $\mathbb{R}^n$ into itself is orthogonal if and only if it preserves the dot product, or (equivalently) if and only if its standard matrix representation is orthogonal, or (equivalently) if and only if it maps unit vectors into unit vectors.
4. The eigenspaces of a symmetric matrix A are mutually orthogonal, and A has n mutually perpendicular eigenvectors.
5. A symmetric matrix A is diagonalizable by an orthogonal matrix C . That is, there exists an orthogonal matrix C such that $D = C^{-1}AC$ is a diagonal matrix.

EXERCISES

In Exercises 1–4, verify that the given matrix is orthogonal, and find its inverse.

$$1. (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \quad 2. \begin{bmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ -\frac{4}{5} & 0 & \frac{3}{5} \\ 0 & 1 & 0 \end{bmatrix}$$

$$3. \frac{1}{7} \begin{bmatrix} 2 & -3 & 6 \\ 3 & 6 & 2 \\ -6 & 2 & 3 \end{bmatrix} \quad 4. \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

If A and D are square matrices, D is diagonal, and AD is orthogonal, then $(AD)^{-1} = (AD)^T$ and $D^{-1}A^{-1} = D^T A^T$ so that $A^{-1} = DD^T A^T = D^2 A^T$. In Exercises 5–8, find the inverse of each matrix A by first finding a diagonal matrix D so that AD has column vectors of length 1, and then applying the formula $A^{-1} = D^2 A^T$.

$$5. \begin{bmatrix} 1 & 3 \\ -1 & 3 \end{bmatrix} \quad 6. \begin{bmatrix} 3 & 0 & 8 \\ -4 & 0 & 6 \\ 0 & 1 & 0 \end{bmatrix}$$

$$7. \begin{bmatrix} 4 & -3 & 6 \\ 6 & 6 & 2 \\ -12 & 2 & 3 \end{bmatrix} \quad 8. \begin{bmatrix} 2 & -1 & 3 & 1 \\ -2 & 1 & 3 & 1 \\ 2 & 1 & -3 & 1 \\ 2 & 1 & 3 & -1 \end{bmatrix}$$

9. Supply a third column vector so that the matrix

$$\begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \end{bmatrix} \text{ is orthogonal.}$$

10. Repeat Exercise 9 for the matrix

$$\begin{bmatrix} \frac{2}{7} & 3/\sqrt{13} \\ \frac{3}{7} & -2/\sqrt{13} \\ \frac{6}{7} & 0 \end{bmatrix}.$$

11. Let (a_1, a_2, a_3) be an ordered orthonormal basis for $\mathbb{R}^3$, and let b be a unit vector with coordinate vector $\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & c \end{bmatrix}$ relative to this basis. Find all possible values for c .

12. Let (a_1, a_2, a_3, a_4) be an ordered orthonormal basis for $\mathbb{R}^4$, and let $[2, 1, 4, -3]$ be the coordinate vector of a vector b in $\mathbb{R}^4$ relative to this basis. Find $\|b\|$.

In Exercises 13–18, find a matrix C such that $D = C^{-1}AC$ is an orthogonal diagonalization of the given symmetric matrix A .

$$13. \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$14. \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}$$

$$15. \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$16. \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$17. \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & -2 & 2 & 1 \\ 1 & 2 & -2 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$18. \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

19. Mark each of the following True or False.
- a. A square matrix is orthogonal if its column vectors are orthogonal.
 - b. Every orthogonal matrix has nullspace $\{0\}$.
 - c. If A^T is orthogonal, then A is orthogonal.
 - d. If A is an $n \times n$ symmetric orthogonal matrix, then $A^2 = I$.
 - e. If A is an $n \times n$ symmetric matrix such that $A^2 = I$, then A is orthogonal.
 - f. If A and B are orthogonal $n \times n$ matrices, then AB is orthogonal.
 - g. Every orthogonal linear transformation carries every unit vector into a unit vector.
 - h. Every linear transformation that carries each unit vector into a unit vector is orthogonal.
 - i. Every map of the plane into itself that is an *isometry* (that is, preserves distance between points) is given by an orthogonal linear transformation.
 - j. Every map of the plane into itself that is an *isometry* and that leaves the origin fixed is given by an orthogonal linear transformation.

20. Let A be an orthogonal $n \times n$ matrix. Show that $\|Ax\| = \|A^{-1}x\|$ for any vector x in $\mathbb{R}^n$.
21. Let A be an orthogonal matrix. Show that A^2 is an orthogonal matrix, too.
22. Show that, if A is an orthogonal matrix, then $\det(A) = \pm 1$.
23. Find a 2×2 matrix with determinant 1 that is not an orthogonal matrix.
24. Let $D = C^{-1}AC$ be a diagonal matrix, where C is an orthogonal matrix. Show that A is symmetric.
25. Let A be an $n \times n$ matrix such that $Ax \cdot Ay = x \cdot y$ for all vectors x and y in $\mathbb{R}^n$. Show that A is an orthogonal matrix.
26. Let A be an $n \times n$ matrix such that $\|Ax\| = \|x\|$ for all vectors x in $\mathbb{R}^n$. Show that A is an orthogonal matrix. [HINT: Show that $x \cdot y = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$, and then use Exercise 25.]
27. Show that the real eigenvalues of an orthogonal matrix must be equal to 1 or -1 . [HINT: Think in terms of linear transformations.]
28. Describe all real diagonal orthogonal matrices.
29.
 - a. Show that a row-interchange elementary matrix is orthogonal.
 - b. Let A be a matrix obtained by permuting (that is, changing the order of) the rows of the $n \times n$ identity matrix. Show that A is an orthogonal matrix.
30. Let $\{a_1, a_2, \dots, a_n\}$ be an orthonormal basis of column vectors for $\mathbb{R}^n$, and let C be an orthogonal $n \times n$ matrix. Show that

$$\{Ca_1, Ca_2, \dots, Ca_n\}$$

is also an orthonormal basis for $\mathbb{R}^n$.

31. Let A and C be orthogonal $n \times n$ matrices. Show that $C^{-1}AC$ is orthogonal.

In Exercises 32–37, determine whether the given linear transformation is orthogonal.

32. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [y, x]$
33. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x, y, z]) = [x, y, 0]$
34. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [2x, y]$
35. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [x, -y]$
36. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [x/2 - \sqrt{3}y/2, -\sqrt{3}x/2 + y/2]$
37. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x, y, z]) = [x/3 + 2y/3 + 2z/3, -2x/3 - y/3 + 2z/3, -2x/3 + 2y/3 - z/3]$
38. Find a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ that preserves the angle between vectors but is not an orthogonal transformation.
39. Show that every 2×2 orthogonal matrix is of one of two forms: either

$$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix},$$

for some angle θ .

40. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation. Show that T is orthogonal if and only if T maps unit vectors to unit vectors. [HINT: Use Exercise 26.]
41. (Real Householder matrix) Let v be a nonzero column vector in $\mathbb{R}^n$. Show that $C_v = I - \frac{2}{v \cdot v}(vv^T)$ is an orthogonal matrix. (These Householder matrices can be used to perform certain important stable reductions of matrices.)

6.4

THE PROJECTION MATRIX

Let W be a subspace of $\mathbb{R}^n$. Projection of vectors in $\mathbb{R}^n$ on W gives a mapping T of $\mathbb{R}^n$ into itself. Figure 6.8 illustrates the projection $T(a + b)$ of $a + b$ on W , and Figure 6.9 illustrates the projection $T(ra)$ of ra on W . These figures suggest that

$$T(a + b) = T(a) + T(b)$$

and

$$T(ra) = rT(a).$$

Thus we expect that T is a linear transformation. If this is the case, there must be a matrix P such that $T(x) = Px$ —namely, the standard matrix representation of T . Recall that in Section 6.1 we showed how to find the projection of a vector b on W by finding a basis for W , then finding a basis for its orthogonal complement $W^\perp$, then finding coordinates of b with respect to the resulting basis of $\mathbb{R}^n$, and so on. It is a somewhat involved process. It would be nice to have a matrix P such that projection of b on W could be found by computing Pb .

In this section, we start by showing that there is indeed a matrix P such that the projection of b on W is equal to Pb . Of course, this then shows that projection is a linear transformation, because we know that multiplication of column vectors in $\mathbb{R}^n$ on the left by any $n \times n$ matrix A gives a linear transformation of $\mathbb{R}^n$ into itself. We derive a formula for this *projection matrix* P corresponding to a subspace W . The formula involves choosing a basis for W , but the final matrix P obtained is independent of the basis chosen.

We have seen that projection on a subspace W is less difficult to compute when an orthonormal basis for W is used. The formula for the matrix P also becomes quite simple when an orthonormal basis is used, and consequently projection becomes an easy computation. Section 6.5 gives an application of these ideas to data analysis.

We will need the following result concerning the rank of $A^T A$ for our work in this section.

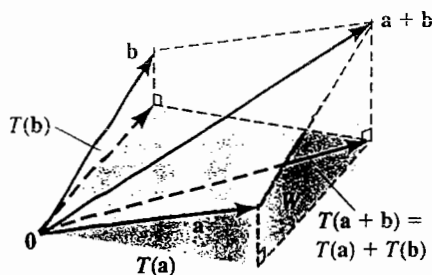


FIGURE 6.8
Projection of $a + b$ on W .

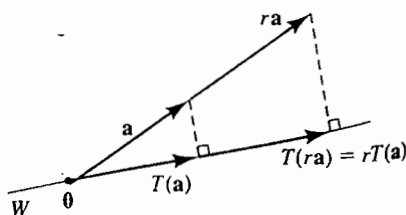


FIGURE 6.9
Projection of ra on W .

THEOREM 6.10 The Rank of $(A^T)A$

Let A be an $m \times n$ matrix of rank r . Then the $n \times n$ symmetric matrix $(A^T)A$ also has rank r .

PROOF We will work with nullspaces. If $\mathbf{v}$ is any solution vector of the system $A\mathbf{x} = \mathbf{0}$, so that $A\mathbf{v} = \mathbf{0}$, then upon multiplying the sides of this last equation on the left by A^T , we see that $\mathbf{v}$ is also a solution of the system $(A^T)A\mathbf{x} = \mathbf{0}$. Conversely, assuming that $(A^T)A\mathbf{w} = \mathbf{0}$ for an $n \times 1$ vector $\mathbf{w}$, then

$$[0] = \mathbf{w}^T[(A^T)A\mathbf{w}] = (A\mathbf{w})^T(A\mathbf{w}),$$

which may be written $\|A\mathbf{w}\|^2 = 0$. That is, $A\mathbf{w}$ is a vector of magnitude 0, and consequently $A\mathbf{w} = \mathbf{0}$. It follows that A and $(A^T)A$ have the same nullspace. Because they have the same number of columns, we apply the rank equation (Theorem 2.5) and conclude that $\text{rank}(A) = \text{rank}((A^T)A)$. $\blacktriangle$

The Formula for the Projection Matrix

Let $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k)$ be a subspace of $\mathbb{R}^n$, where the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ are independent. Let $\mathbf{b}$ be a vector in $\mathbb{R}^n$, and let $\mathbf{p} = \mathbf{b}_W$ be its projection on W . From the unique decomposition

$$\mathbf{b} = \mathbf{p} + (\mathbf{b} - \mathbf{p})$$

$$\mathbf{b}_W \quad \mathbf{b}_{W^\perp}$$

explained in Section 6.1, we see that the projection vector $\mathbf{p}$ is the unique vector satisfying the following two properties. (See Figure 6.10 for an illustration when $\dim(W) = 2$.)

Properties of the Projection $\mathbf{p}$ of Vector $\mathbf{b}$ on the Subspace W

1. The vector $\mathbf{p}$ must lie in the subspace W .
2. The vector $\mathbf{b} - \mathbf{p}$ must be perpendicular to every vector in W .

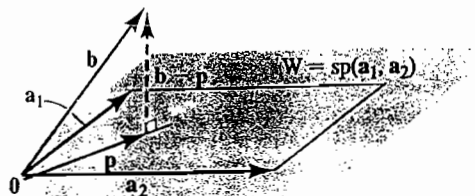


FIGURE 6.10
The projection of $\mathbf{b}$ on $\text{sp}(\mathbf{a}_1, \mathbf{a}_2)$.

If we write our vectors in $\mathbb{R}^n$ as column vectors, the subspace W is the column space of the $n \times k$ matrix A , whose columns are $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$. All vectors in $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k)$ have the form $A\mathbf{x}$, where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix} \quad \text{for any scalars } x_1, x_2, \dots, x_k.$$

Because $\mathbf{p}$ lies in the space W , we see that $\mathbf{p} = A\mathbf{r}$, where

$$\mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_k \end{bmatrix} \quad \text{for some scalars } r_1, r_2, \dots, r_k.$$

Because $\mathbf{b} - A\mathbf{r}$ must be perpendicular to each vector in W , the dot product of $\mathbf{b} - A\mathbf{r}$ and $A\mathbf{x}$ must be zero for all vectors $\mathbf{x}$. This dot-product condition can be written as the matrix equation

$$(A\mathbf{x})^T(\mathbf{b} - A\mathbf{r}) = \mathbf{x}^T(A^T\mathbf{b} - A^T A\mathbf{r}) = [0].$$

In other words, the dot product of the vectors $\mathbf{x}$ and $A^T\mathbf{b} - A^T A\mathbf{r}$ must be zero for all vectors $\mathbf{x}$. This can happen only if the vector $A^T\mathbf{b} - A^T A\mathbf{r}$ is itself the zero vector. (See Exercise 41 in Section 1.3.) Therefore, we have

$$A^T\mathbf{b} - A^T A\mathbf{r} = \mathbf{0}. \quad (1)$$

Now the $k \times k$ matrix $A^T A$ appearing in Eq. (1) is invertible, because it has the same rank as A (see Theorem 6.10); and A has rank k , because its columns are independent. Solving Eq. (1) for $\mathbf{r}$, we obtain

$$\mathbf{r} = (A^T A)^{-1} A^T \mathbf{b}. \quad (2)$$

Denoting the projection $\mathbf{p} = A\mathbf{r}$ of $\mathbf{b}$ on W by $\mathbf{p} = \mathbf{b}_W$ and writing $\mathbf{b}$ as a column vector, we obtain the formula in the following box.

Projection $\mathbf{b}_W$ of $\mathbf{b}$ on the Subspace W

Let $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k)$ be a k -dimensional subspace of $\mathbb{R}^n$, and let A have as columns the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$. The projection of $\mathbf{b}$ in $\mathbb{R}^n$ on W is given by

$$\mathbf{b}_W = A(A^T A)^{-1} A^T \mathbf{b}. \quad (3)$$

We leave as Exercise 12 the demonstration that the formula in Section 6.1 for projecting $\mathbf{b}$ on $\text{sp}(\mathbf{a})$ can be written in the form of formula (3), using matrices of appropriate size.

Our first example reworks Example 1 in Section 6.1, using formula (3).

EXAMPLE 1 Using formula (3), find the projection of the vector $\mathbf{b}$ on the subspace $W = \text{sp}(\mathbf{a})$, where

$$\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \text{and} \quad \mathbf{a} = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}.$$

SOLUTION Let A be the matrix whose single column is $\mathbf{a}$. Then

$$A^T A = \begin{bmatrix} 2 & 4 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix} = [29].$$

Putting this into formula (3), we have

$$\mathbf{b}_W = A(A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix} \left(\frac{1}{29} \right) \begin{bmatrix} 2 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \left(\frac{1}{29} \right) \begin{bmatrix} 4 & 8 & 6 \\ 8 & 16 & 12 \\ 6 & 12 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \left(\frac{19}{29} \right) \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}.$$

We refer to the matrix $A(A^T A)^{-1} A^T$ in formula (3) as the **projection matrix** for the subspace W . It takes any vector $\mathbf{b}$ in $\mathbb{R}^n$ and, by left multiplication, projects it onto the vector $\mathbf{b}_W$, which lies in W . We will show shortly that this matrix is uniquely determined by W , which allows us to use the definite article *the* when talking about it. Before demonstrating this uniqueness, we box the formula for this matrix and present two more examples.

The Projection Matrix P for the Subspace W

Let $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k)$ be a k -dimensional subspace of $\mathbb{R}^n$, and let A have as columns the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$. The projection matrix for the subspace W is given by

$$P = A(A^T A)^{-1} A^T.$$

EXAMPLE 2 Find the projection matrix for the x_2, x_3 -plane in $\mathbb{R}^3$.

SOLUTION The x_2, x_3 -plane is the subspace $W = \text{sp}(\mathbf{e}_2, \mathbf{e}_3)$, where $\mathbf{e}_2$ and $\mathbf{e}_3$ are the column vectors of the matrix

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We find that $A^T A = I$, the 2×2 identity matrix, and that the projection matrix is

$$P = A(A^T A)^{-1} A^T = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Notice that P projects each vector

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \text{ in } \mathbb{R}^3 \text{ onto } P \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 0 \\ b_2 \\ b_3 \end{bmatrix}$$

in the x_2, x_3 -plane, as expected. ■

EXAMPLE 3 Find the matrix that projects vectors in $\mathbb{R}^3$ on the plane $2x - y - 3z = 0$. Also, find the projection of a general vector $\mathbf{b}$ in $\mathbb{R}^3$ on this plane.

SOLUTION We observe that the given plane contains the zero vector and can therefore be written as the subspace $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$, where $\mathbf{a}_1$ and $\mathbf{a}_2$ are any two nonzero and nonparallel vectors in the plane. We choose

$$\mathbf{a}_1 = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix} \text{ and } \mathbf{a}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix},$$

so that

$$A = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -1 & 0 \end{bmatrix}.$$

Then

$$(A^T A)^{-1} = \begin{bmatrix} 10 & 6 \\ 6 & 5 \end{bmatrix}^{-1} = \frac{1}{14} \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix}$$

and the desired matrix is

$$\begin{aligned} P &= \frac{1}{14} \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & -6 \\ -6 & 10 \end{bmatrix} \begin{bmatrix} 0 & 3 & -1 \\ 1 & 2 & 0 \end{bmatrix} \\ &= \frac{1}{14} \begin{bmatrix} -6 & 10 \\ 3 & 2 \\ -5 & 6 \end{bmatrix} \begin{bmatrix} 0 & 3 & -1 \\ 1 & 2 & 0 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 10 & 2 & 6 \\ 2 & 13 & -3 \\ 6 & -3 & 5 \end{bmatrix}. \end{aligned}$$

Each vector $\mathbf{b} = [b_1, b_2, b_3]$ in $\mathbb{R}^3$ projects onto the vector

$$\mathbf{b}_W = P\mathbf{b} = \frac{1}{14} \begin{bmatrix} 10 & 2 & 6 \\ 2 & 13 & -3 \\ 6 & -3 & 5 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 10b_1 + 2b_2 + 6b_3 \\ 2b_1 + 13b_2 - 3b_3 \\ 6b_1 - 3b_2 + 5b_3 \end{bmatrix}.$$

Uniqueness and Properties of the Projection Matrix

It might appear from the formula $P = A(A^T A)^{-1} A^T$ that a projection matrix P for the subspace W of $\mathbb{R}^n$ depends on the particular choice of basis for W that is used for the column vectors of A . However, this is not so. Because the projection map of $\mathbb{R}^n$ onto W can be computed for $\mathbf{x} \in \mathbb{R}^n$ by the multiplication $P\mathbf{x}$ of the vector $\mathbf{x}$ on the left by the matrix P , we see that this projection map is a linear transformation, and P must be its *unique* standard matrix representation. Thus we may refer to the matrix $P = A(A^T A)^{-1} A^T$ as *the* projection matrix for the subspace W . We summarize our work in a theorem.

THEOREM 6.11 Projection Matrix

Let W be a subspace of $\mathbb{R}^n$. There is a unique $n \times n$ matrix P such that, for each column vector $\mathbf{b}$ in $\mathbb{R}^n$, the vector $P\mathbf{b}$ is the projection of $\mathbf{b}$ on W . This projection matrix P can be found by selecting any basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ for W and computing $P = A(A^T A)^{-1} A^T$, where A is the $n \times k$ matrix having column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$.

Exercise 16 indicates that the projection matrix P given in Theorem 6.11 satisfies two properties:

Properties of a Projection Matrix P

- | | |
|----------------|--------------------|
| 1. $P^2 = P$. | P is idempotent. |
| 2. $P^T = P$. | P is symmetric. |

We can use property 2 as a partial check for errors in long computations that lead to P , as in Example 3. These two properties completely characterize the projection matrices, as we now show.

THEOREM 6.12 Characterization of Projection Matrices

The projection matrix P for a subspace W of $\mathbb{R}^n$ is both idempotent and symmetric. Conversely, every $n \times n$ matrix that is both idempotent and symmetric is a projection matrix: specifically, it is the projection matrix for its column space.

PROOF Exercise 16 indicates that a projection matrix is both idempotent and symmetric.

To establish the converse, let P be an $n \times n$ matrix that is both symmetric and idempotent. We show that P is the projection matrix for its own column space W . Let $\mathbf{b}$ be any vector in $\mathbb{R}^n$. By Theorem 6.11, we need show only that

$P\mathbf{b}$ satisfies the characterizing properties of the projection of $\mathbf{b}$ on W given in the box on page 361. Now $P\mathbf{b}$ surely lies in the column space W of P , because W consists of all vectors $P\mathbf{x}$ for any vector $\mathbf{x}$ in $\mathbb{R}^n$. The second requirement is that $\mathbf{b} - P\mathbf{b}$ must be perpendicular to each vector $P\mathbf{x}$ in the column space of P . Writing the dot product of $\mathbf{b} - P\mathbf{b}$ and $P\mathbf{x}$ in matrix form, and using the hypotheses $P^2 = P = P^T$, we have

$$\begin{aligned}(\mathbf{b} - P\mathbf{b})^T P\mathbf{x} &= ((I - P)\mathbf{b})^T P\mathbf{x} = \mathbf{b}^T(I - P)^T P\mathbf{x} \\&= \mathbf{b}^T(I - P)P\mathbf{x} = \mathbf{b}^T(P - P^2)\mathbf{x} \\&= \mathbf{b}^T(P - P)\mathbf{x} = \mathbf{b}^T \mathbf{0} = [0].\end{aligned}$$

Because their dot product is zero, we see that $\mathbf{b} - P\mathbf{b}$ and $P\mathbf{x}$ are indeed perpendicular, and our proof is complete. $\blacktriangle$

The Orthonormal Case

In Example 2, we saw that the projection matrix for the x_2, x_3 -plane in $\mathbb{R}^3$ has the very simple description

$$P = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The usually complicated formula $P = A(A^T A)^{-1} A^T$ was simplified when we used the standard unit coordinate vectors in our basis for W . Namely, we had

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{so} \quad A^T A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Thus, $(A^T A)^{-1} = I$, too, which simplifies what is normally the worst part of the computation in our formula for P . This simplification can be made in computing P for any subspace W of $\mathbb{R}^n$, provided that we know an *orthonormal basis* $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ for W . If A is the $n \times k$ matrix whose columns are $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$, then we know that $A^T A = I$. The formula for the projection matrix becomes

$$P = A(A^T A)^{-1} A^T = A I A^T = A A^T.$$

We box this result for easy reference.

Projection Matrix: Orthonormal Case

Let $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ be an orthonormal basis for a subspace W of $\mathbb{R}^n$. The projection matrix for W is

$$P = A A^T, \quad (4)$$

where A is the $n \times k$ matrix having column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$.

EXAMPLE 4 Find the projection matrix for the subspace $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ of $\mathbb{R}^3$ if

$$\mathbf{a}_1 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \quad \text{and} \quad \mathbf{a}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$

Find the projection of each vector $\mathbf{b}$ in $\mathbb{R}^3$ on W .

SOLUTION Let

$$A = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} \\ -1/\sqrt{3} & 1/\sqrt{2} \\ 1/\sqrt{3} & 0 \end{bmatrix}.$$

Then

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

so

$$P = AA^T = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{bmatrix}.$$

Each column vector $\mathbf{b}$ in $\mathbb{R}^3$ projects onto

$$\mathbf{b}_W = P\mathbf{b} = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5b_1 + b_2 + 2b_3 \\ b_1 + 5b_2 - 2b_3 \\ 2b_1 - 2b_2 + 2b_3 \end{bmatrix}.$$

Alternatively, we can compute $\mathbf{b}_W$ directly, using boxed Eq. (4) in Section 6.2. We obtain

$$\begin{aligned} \mathbf{b}_W &= (\mathbf{b} \cdot \mathbf{a}_1)\mathbf{a}_1 + (\mathbf{b} \cdot \mathbf{a}_2)\mathbf{a}_2 = \frac{1}{3} \begin{bmatrix} b_1 - b_2 + b_3 \\ -b_1 + b_2 - b_3 \\ b_1 - b_2 + b_3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} b_1 + b_2 \\ b_1 + b_2 \\ 0 \end{bmatrix} \\ &= \frac{1}{6} \begin{bmatrix} 5b_1 + b_2 + 2b_3 \\ b_1 + 5b_2 - 2b_3 \\ 2b_1 - 2b_2 + 2b_3 \end{bmatrix}. \end{aligned}$$

SUMMARY

Let $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ be a basis for a subspace W of $\mathbb{R}^n$, and let A be the $n \times k$ matrix having $\mathbf{a}_j$ as j th column vector, so that W is the column space of A .

1. The projection of a column vector $\mathbf{b}$ in $\mathbb{R}^n$ on W is $\mathbf{b}_W = A(A^T A)^{-1} A^T \mathbf{b}$.
2. The matrix $P = A(A^T A)^{-1} A^T$ is the projection matrix for the subspace W . It is the unique matrix P such that, for every vector $\mathbf{b}$ in $\mathbb{R}^n$, the vector $P\mathbf{b}$ lies in W and the vector $\mathbf{b} - P\mathbf{b}$ is perpendicular to every vector in W .
3. If the basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k\}$ of the subspace W is orthonormal, the projection matrix for W is $P = AA^T$.

4. The projection matrix P of a subspace W is idempotent and symmetric. Every symmetric idempotent matrix is the projection matrix for its column space.

EXERCISES

In Exercises 1–8, find the projection matrix for the given subspace, and find the projection of the indicated vector on the subspace.

- [1, 2, 1] on $\text{sp}([2, 1, -1])$ in $\mathbb{R}^3$
- [1, 3, 4] on $\text{sp}([1, -1, 2])$ in $\mathbb{R}^3$
- [2, -1, 3] on $\text{sp}([2, 1, 1], [-1, 2, 1])$ in $\mathbb{R}^3$
- [1, 2, 1] on $\text{sp}([3, 0, 1], [1, 1, 1])$ in $\mathbb{R}^3$
- [1, 3, 1] on the plane $x + y - 2z = 0$ in $\mathbb{R}^3$
- [4, 2, -1] on the plane $3x + 2y + z = 0$ in $\mathbb{R}^3$
- [1, 2, 1, 3] on $\text{sp}([1, 2, 1, 1], [-1, 1, 0, -1])$ in $\mathbb{R}^4$
- [1, 1, 2, 1] on $\text{sp}([1, 1, 1, 1], [1, -1, 1, -1], [-1, 1, 1, -1])$ in $\mathbb{R}^4$
- Find the projection matrix for the x_1, x_2 -plane in $\mathbb{R}^3$.
- Find the projection matrix for the x_1, x_3 -coordinate subspace of $\mathbb{R}^4$.
- Find the projection matrix for the x_1, x_2, x_4 -coordinate subspace of $\mathbb{R}^4$.
- Show that boxed Eq. (3) of this section reduces to Eq. (1) of Section 6.1 for projecting $\mathbf{b}$ on $\text{sp}(\mathbf{a})$.
- Give a geometric argument indicating that every projection matrix is idempotent.
- Let $\mathbf{a}$ be a unit column vector in $\mathbb{R}^n$. Show that $\mathbf{a}\mathbf{a}^T$ is the projection matrix for the subspace $\text{sp}(\mathbf{a})$.
- Mark each of the following True or False.
 - a. A subspace W of dimension k in $\mathbb{R}^n$ has associated with it a $k \times k$ projection matrix.
 - b. Every subspace W of $\mathbb{R}^n$ has associated with it an $n \times n$ projection matrix.
 - c. Projection of $\mathbb{R}^n$ on a subspace W is a linear transformation of $\mathbb{R}^n$ into itself.
 - d. Two different subspaces of $\mathbb{R}^n$ may have the same projection matrix.
 - e. Two different matrices may be projection matrices for the same subspace of $\mathbb{R}^n$.
 - f. Every projection matrix is symmetric.
 - g. Every symmetric matrix is a projection matrix.
 - h. An $n \times n$ symmetric matrix A is a projection matrix if and only if $A^2 = A$.
 - i. Every symmetric idempotent matrix is the projection matrix for its column space.
 - j. Every symmetric idempotent matrix is the projection matrix for its row space.
- Show that the projection matrix $P = A(A^T A)^{-1}A^T$ given in Theorem 6.11 satisfies the following two conditions:
 - a. $P^2 = P$,
 - b. $P^T = P$.
- What is the projection matrix for the subspace $\mathbb{R}^n$ of $\mathbb{R}^n$?
- Let U be a subspace of W , which is a subspace of $\mathbb{R}^n$. Let P be the projection matrix for W , and let R be the projection matrix for U . Find PR and RP . [Hint: Argue geometrically.]
- Let P be the projection matrix for a k -dimensional subspace of $\mathbb{R}^n$.
 - a. Find all eigenvalues of P .
 - b. Find the algebraic multiplicity and the geometric multiplicity of each eigenvalue found in part (a).
 - c. Explain how we can deduce that P is diagonalizable, without using the fact that P is a symmetric matrix.
- Show that every symmetric matrix whose only eigenvalues are 0 and 1 is a projection matrix.
- Find all invertible projection matrices.

In Exercises 22–28, find the projection matrix for the subspace W having the given orthonormal basis. The vectors are given in row notation to save space in printing.

- $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ in $\mathbb{R}^3$, where
 - $\mathbf{a}_1 = [1/\sqrt{2}, 0, -1/\sqrt{2}]$ and
 - $\mathbf{a}_2 = [1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}]$
- $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ in $\mathbb{R}^3$, where
 - $\mathbf{a}_1 = [\frac{3}{5}, \frac{4}{5}, 0]$ and $\mathbf{a}_2 = [0, 0, 1]$

24. $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ in $\mathbb{R}^4$, where
 $\mathbf{a}_1 = \left[\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}}, \frac{1}{2}, -\frac{1}{2} \right]$ and
 $\mathbf{a}_2 = \left[\frac{4}{\sqrt{5}}, -\frac{3}{\sqrt{5}}, \frac{1}{2}, \frac{1}{2} \right]$
25. $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ in $\mathbb{R}^4$, where $\mathbf{a}_1 = \left[\frac{2}{7}, 0, \frac{3}{7}, -\frac{6}{7} \right]$
and $\mathbf{a}_2 = \left[-\frac{3}{7}, \frac{6}{7}, \frac{2}{7}, 0 \right]$
26. $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ in $\mathbb{R}^4$, where
 $\mathbf{a}_1 = \left[0, -\frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right]$ and $\mathbf{a}_2 = \left[\frac{2}{3}, 0, -\frac{1}{3}, \frac{2}{3} \right]$
27. $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ in $\mathbb{R}^4$, where
 $\mathbf{a}_1 = [1/\sqrt{3}, 0, 1/\sqrt{3}, 1/\sqrt{3}]$,
 $\mathbf{a}_2 = [1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3}, 0]$, and
 $\mathbf{a}_3 = [1/\sqrt{3}, -1/\sqrt{3}, 0, -1/\sqrt{3}]$
28. $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ in $\mathbb{R}^4$, where
 $\mathbf{a}_1 = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right]$, $\mathbf{a}_2 = \left[-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right]$, and
 $\mathbf{a}_3 = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2} \right]$

In Exercises 29–32, find the projection of $\mathbf{b}$ on W .

29. The subspace W in Exercise 22;
 $\mathbf{b} = [6, -12, -6]$
30. The subspace W in Exercise 23;
 $\mathbf{b} = [20, -15, 5]$
31. The subspace W in Exercise 26;
 $\mathbf{b} = [9, 0, -9, 18]$
32. The subspace W in Exercise 28;
 $\mathbf{b} = [4, -12, -4, 0]$
33. Let W be a subspace of $\mathbb{R}^n$, and let P be the projection matrix for W . **Reflection of $\mathbb{R}^n$ in W** is the mapping of $\mathbb{R}^n$ into itself that carries each vector $\mathbf{b}$ in $\mathbb{R}^n$ into its reflection $\mathbf{b}_r$ according to the following geometric description:

Let $\mathbf{p}$ be the projection of $\mathbf{b}$ on W . Starting at the tip of $\mathbf{b}$, travel in a straight line to the tip of $\mathbf{p}$, and then continue in the same

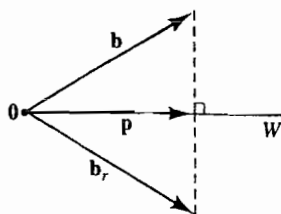


FIGURE 6.11
Reflection of $\mathbb{R}^n$ through W .

direction an equal distance to arrive at $\mathbf{b}_r$.
(See Figure 6.11.)

Show that $\mathbf{b}_r = (2P - I)\mathbf{b}$. (Notice that, because reflection can be accomplished by matrix multiplication, this reflection must be a linear transformation of $\mathbb{R}^n$ into itself.)

 The formula $A(A^T A)^{-1} A^T$ for a projection matrix can be tedious to compute using pencil and paper, but the routine **MATCOMP** in **LINTEK**, or **MATLAB**, can do it easily. In Exercises 34–38, use **MATCOMP** or **MATLAB** to find the indicated vector projections.

34. The projections in $\mathbb{R}^6$ of $[-1, 2, 3, 1, 6, 2]$
and $[2, 0, 3, -1, 4, 5]$ on
 $\text{sp}([1, -2, 3, 1, 4, 0])$
35. The projections in $\mathbb{R}^3$ of $[1, -1, 4]$,
 $[3, 3, -1]$, and $[-2, 4, 7]$ on
 $\text{sp}([1, 3, -4], [2, 0, 3])$
36. The projections in $\mathbb{R}^4$ of $[-1, 3, 2, 0]$ and
 $[4, -1, 1, 5]$ on $\text{sp}([0, 1, 2, 1], [-1, 2, 1, 4])$
37. The projections in $\mathbb{R}^4$ of $[2, 1, 0, 3]$,
 $[1, 1, -1, 2]$, and $[4, 3, 1, 3]$ on
 $\text{sp}([1, 0, -1, 0], [1, 2, -1, 4], [2, 1, 3, -1])$
38. The projections in $\mathbb{R}^5$ of $[2, 1, -3, 2, 4]$ and
 $[1, -4, 0, 1, 5]$ on
 $\text{sp}([3, 1, 4, 0, 1], [2, 1, 3, -5, 1])$

6.5

THE METHOD OF LEAST SQUARES

The Nature of the Problem

In this section we apply our work on projections to problems of data analysis. Suppose that data measurements of the form $(\bar{a}_i, b_i)$ are obtained from

测量. 修正

observation or experimentation and are plotted as data points in the x, y -plane. It is desirable to find a mathematical relationship $y = f(x)$ that represents the data reasonably well, so that we can make predictions of data values that were not measured. Geometrically, this means that we would like the graph of $y = f(x)$ in the plane to pass very close to our data points. Depending on the nature of the experiment and the configuration of the plotted data points, we might decide on an appropriate type of function $y = f(x)$ such as a linear function, a quadratic function, or an exponential function. We illustrate with three types of problems.

- PROBLEM 1 According to Hooke's law, the distance that a spring stretches is proportional to the force applied. Suppose that we attach four different weights a_1, a_2, a_3 , and a_4 in turn to the bottom of a spring suspended vertically. We measure the four lengths b_1, b_2, b_3 , and b_4 of the stretched spring, and the data in Table 6.1 are obtained. Because of Hooke's law, we expect the data points (a_i, b_i) to be close to some line with equation

$$y = f(x) = r_0 + r_1x,$$

where r_0 is the unstretched length of the spring and r_1 is the *spring constant*. That is, if our measurements were exact and the spring ideal, we would have $b_i = r_0 + r_1a_i$ for specific values r_0 and r_1 . ■

TABLE 6.1

a_i = Weight in ounces	2.0	4.0	5.0	6.0
b_i = Length in inches	6.5	8.5	11.0	12.5

In Problem 1, we have only the two unknowns r_0 and r_1 ; and in theory, just two measurements should suffice to find them. In practice, however, we expect to have some error in physical measurements. It is standard procedure to make more measurements than are theoretically necessary in the hope that the errors will roughly cancel each other out, in accordance with the laws of probability. Substitution of each data point (a_i, b_i) from Problem 1 into the equation $y = r_0 + r_1x$ gives a single linear equation in the two unknowns r_0 and r_1 . The four data points of Problem 1 thus give rise to a linear system of four equations in only two unknowns. Such a linear system with more equations than unknowns is called **overdetermined**, and one expects to find that the system is *inconsistent*, having no actual solution. It will be our task to find values for the unknowns r_0 and r_1 that will come as close as possible, in some sense, to satisfying all four of the equations.

We have used the illustration presented in Problem 1 to introduce our goal in this section, and we will solve the problem in a moment. We first present two more hypothetical problems, which we will also solve later in the section.

PROBLEM 2 At a recent boat show, the observations listed in Table 6.2 were made relating the prices b_i of sailboats and their weights a_i . Plotting the data points (a_i, b_i) , as shown in Figure 6.12, we might expect a quadratic function of the form

$$y = f(x) = r_0 + r_1x + r_2x^2$$

to fit the data fairly well. ■

TABLE 6.2

a_i = Weight in tons	2	4	5	8
b_i = Price in units of \$10,000	1	3	5	12

PROBLEM 3 A population of rabbits on a large island was estimated each year from 1991 to 1994, giving the data in Table 6.3. Knowing that population growth is exponential in the absence of disease, predators, famine, and so on, we expect an exponential function

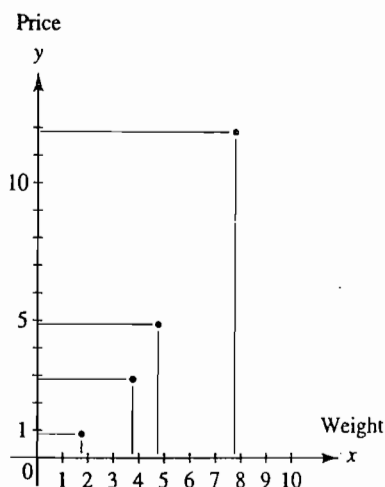
$$y = f(x) = re^{sx}$$

to provide the best representation of these data. Notice that, by using logarithms, we can convert this exponential function into a form linear in x :

$$\ln y = \ln r + sx.$$

TABLE 6.3

a_i = (Year observed) - 1990	1	2	3	4
b_i = Number of rabbits in units of 1000	3	4.5	8	17

FIGURE 6.12
Problem 2 data.

The Method of Least Squares

Consider now the problem of finding a linear function $f(x) = r_0 + r_1x$ that best fits data points (a_i, b_i) for $i = 1, 2, \dots, m$, where $m > 2$. Geometrically, this amounts to finding the line in the plane that comes closest, in some sense, to passing through the m data points. If there were no error in our measurements and our data were truly linear, then for some r_0 and r_1 we would have

$$b_i = r_0 + r_1 a_i \quad \text{for } i = 1, 2, \dots, m.$$

These m linear equations in the two unknowns r_0 and r_1 form an *overdetermined* system of equations that probably has no solution. Our data points actually satisfy a system of linear approximations, which can be expressed in matrix form as

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \approx \begin{bmatrix} 1 & a_1 \\ 1 & a_2 \\ \vdots & \vdots \\ 1 & a_m \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \end{bmatrix} \quad (1)$$

$\mathbf{b} \qquad \mathbf{A} \qquad \mathbf{r}$

or simply as $\mathbf{b} \approx \mathbf{A}\mathbf{r}$. We try to find an optimal solution vector $\bar{\mathbf{r}}$ for the system (1) of approximations. For each vector $\mathbf{r}$, the error vector $\mathbf{A}\mathbf{r} - \mathbf{b}$ measures how far our system (1) is from being a system of equations with solution vector $\mathbf{r}$. The absolute values of the components of the vector $\mathbf{A}\mathbf{r} - \mathbf{b}$ represent the vertical distances $d_i = |r_0 + r_1 a_i - b_i|$, shown in Figure 6.13.

We want to minimize, in some sense, our error vector $\mathbf{A}\mathbf{r} - \mathbf{b}$. A number of different methods for minimization are very useful. For example, one might want to minimize the maximum of the distances d_i . We study just one sense of

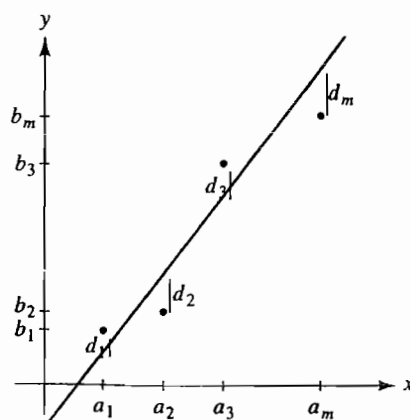


FIGURE 6.13
The distances d_i .

minimization, the one that probably seems most natural at this point. We will find $\mathbf{r} = \bar{\mathbf{r}}$ to minimize the *length* $\|\mathbf{A}\mathbf{r} - \mathbf{b}\|$ of our error vector. Minimizing $\|\mathbf{A}\mathbf{r} - \mathbf{b}\|$ can be achieved by minimizing $\|\mathbf{A}\mathbf{r} - \mathbf{b}\|^2$, which means minimizing the sum

$$d_1^2 + d_2^2 + \cdots + d_m^2$$

of the squares of the distances in Figure 6.13. Hence the name **method of least squares** given to this procedure.

If $\mathbf{a}_1$ and $\mathbf{a}_2$ denote the columns of A in system (1), the vector $\mathbf{A}\mathbf{r} = r_1\mathbf{a}_1 + r_2\mathbf{a}_2$ lies in the column space $W = \text{sp}(\mathbf{a}_1, \mathbf{a}_2)$ of A . From Figure 6.14, we see geometrically that, of all the vectors $\mathbf{A}\mathbf{r}$ in W , the one that minimizes $\|\mathbf{A}\mathbf{r} - \mathbf{b}\|$ is the *projection* $\mathbf{b}_W = \mathbf{A}\bar{\mathbf{r}}$ of $\mathbf{b}$ on W . Recall that we proved this minimization property of $\mathbf{b}_W$ algebraically on page 328. Formula (3) in Section 6.4 shows that then $\mathbf{A}\bar{\mathbf{r}} = (A^T A)^{-1} A^T \mathbf{b}$. Thus our optimal solution vector $\bar{\mathbf{r}}$ is

$$\bar{\mathbf{r}} = (A^T A)^{-1} A^T \mathbf{b}. \quad (2)$$

The 2×2 matrix $A^T A$ is invertible as long as the columns of A are independent, as shown by Theorem 6.10. For our matrix A shown in system (1), this just means that not all the values a_i are the same. Geometrically, this corresponds to saying that our data points in the plane do not all lie on a vertical line.

Note that the equation $\bar{\mathbf{r}} = (A^T A)^{-1} A^T \mathbf{b}$ can be rewritten as $(A^T A)\bar{\mathbf{r}} = A^T \mathbf{b}$. Thus $\bar{\mathbf{r}}$ appears as the unique solution of the consistent linear system

$$(A^T A)\mathbf{r} = A^T \mathbf{b}. \quad (3)$$

Unless the coefficient matrix $A^T A$ is of very small size, it is more efficient when using a computer to solve the linear system (3), reducing an augmented matrix as usual, rather than to invert $A^T A$ and then multiply it by $A^T \mathbf{b}$ as in Eq. (2). In our examples and exercises, $A^T A$ will often be a 2×2 matrix whose inverse can be written down at once by taking $1/\det(A^T A)$ times the adjoint of $A^T A$ (see page 269), making the use of Eq. (2) practical. We summarize in a box, and proceed with examples.

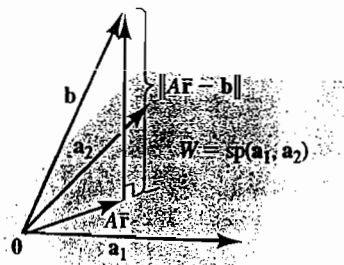


FIGURE 6.14
The length $\|\mathbf{A}\bar{\mathbf{r}} - \mathbf{b}\|$.

Least-Squares Solution of $A\mathbf{r} \approx \mathbf{b}$

Let A be a matrix with independent column vectors. The least-squares solution $\bar{\mathbf{r}}$ of $A\mathbf{r} \approx \mathbf{b}$ can be computed in either of the following ways:

1. Compute $\mathbf{r} = (A^T A)^{-1} A^T \mathbf{b}$.
2. Solve $(A^T A)\mathbf{r} = A^T \mathbf{b}$.

When a computer is being used, the second method is more efficient.

EXAMPLE 1 Find the least-squares fit to the data points in Problem 1 by a straight line—that is, by a linear function $y = r_0 + r_1 x$.

SOLUTION We form the system $\mathbf{b} \approx A\mathbf{r}$ in system (1) for the data in Table 6.1:

$$\underbrace{\begin{bmatrix} 6.5 \\ 8.5 \\ 11.0 \\ 12.5 \end{bmatrix}}_{\mathbf{b}} \approx \underbrace{\begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}}_A \underbrace{\begin{bmatrix} r_0 \\ r_1 \end{bmatrix}}_{\mathbf{r}}.$$

We have

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} 4 & 17 \\ 17 & 81 \end{bmatrix}$$

and

$$(A^T A)^{-1} = \frac{1}{35} \begin{bmatrix} 81 & -17 \\ -17 & 4 \end{bmatrix}$$

From Eq. (3), we obtain the least-squares solution

$$\begin{aligned} \bar{\mathbf{r}} &= (A^T A)^{-1} A^T \mathbf{b} = \frac{1}{35} \begin{bmatrix} 81 & -17 \\ -17 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 6.5 \\ 8.5 \\ 11.0 \\ 12.5 \end{bmatrix} \\ &= \frac{1}{35} \begin{bmatrix} 47 & 13 & -4 & -21 \\ -9 & -1 & 3 & 7 \end{bmatrix} \begin{bmatrix} 6.5 \\ 8.5 \\ 11.0 \\ 12.5 \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 109.5 \\ 53.5 \end{bmatrix} \approx \begin{bmatrix} 3.1 \\ 1.5 \end{bmatrix}. \end{aligned}$$

Therefore, the linear function that best fits the data in the least-squares sense is $y \approx f(x) = 1.5x + 3.1$. This line and the data points are shown in Figure 6.15. ■

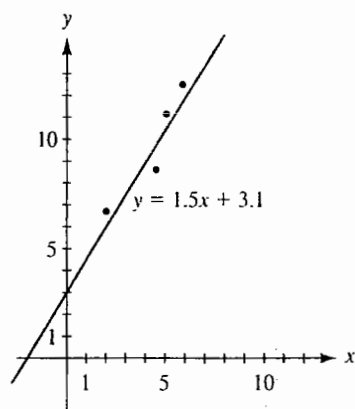


FIGURE 6.15
The least-squares fit of data points.

EXAMPLE 2 Use the method of least squares to fit the data in Problem 3 by an exponential function $y = f(x) = re^{sx}$.

SOLUTION We use logarithms and convert the exponential equation to an equation that is linear in x :

$$\ln y = \ln r + sx.$$

TABLE 6.4

$x = a_i$	1	2	3	4
$y = b_i$	3	4.5	8	17
$z = \ln(b_i)$	1.10	1.50	2.08	2.83

HISTORICAL NOTE A TECHNIQUE VERY CLOSE TO THAT OF LEAST SQUARES was developed by Roger Cotes (1682–1716), the gifted mathematician who edited the second edition of Isaac Newton's *Principia*, in a work dealing with errors in astronomical observations, written around 1715.

The complete principle, however, was first formulated by Carl Gauss at around the age of 16 while he was adjusting approximations in dealing with the distribution of prime numbers. Gauss later stated that he used the method frequently over the years—for example, when he did calculations concerning the orbits of asteroids. Gauss published the method in 1809 and gave a definitive exposition 14 years later.

On the other hand, it was Adrien-Marie Legendre (1752–1833), founder of the theory of elliptic functions, who first published the method of least squares, in an 1806 work on determining the orbits of comets. After Gauss's 1809 publication, Legendre wrote to him, censuring him for claiming the method as his own. Even as late as 1827, Legendre was still berating Gauss for “appropriating the discoveries of others.” In fact, the problem lay in Gauss's failure to publish many of his discoveries promptly; he mentioned them only after they were published by others.

From Table 6.3, we obtain the data in Table 6.4 to use for our logarithmic equation. We obtain

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(A^T A)^{-1} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{5} \end{bmatrix}$$

$$(A^T A)^{-1} A^T = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{3}{10} & -\frac{1}{10} & \frac{1}{10} & \frac{3}{10} \end{bmatrix}.$$

Multiplying this last matrix on the right by

$$\begin{bmatrix} 1.10 \\ 1.50 \\ 2.08 \\ 2.83 \end{bmatrix},$$

we obtain, from Eq. (3),

$$\bar{r} = \begin{bmatrix} \ln r \\ s \end{bmatrix} \approx \begin{bmatrix} .435 \\ .577 \end{bmatrix}.$$

Thus $r = e^{.435} \approx 1.54$, and we obtain $y = \hat{f}(x) = 1.54e^{.577x}$ as a fitting exponential function.

The graph of the function and of the data points in Table 6.5 is shown in Figure 6.16. On the basis of the function $\hat{f}(x)$ obtained, we can project the population of rabbits on the island in the year 2000 to be about

$$\hat{f}(10) \cdot 1000 \approx 494,000 \text{ rabbits,}$$

unless predators, disease, or lack of food interferes. ■

TABLE 6.5

a_i	b_i	$f(a_i)$
1	3	2.7
2	4.5	4.9
3	8	8.7
4	17	15.5

In Example 2, we used the least-squares method with the *logarithm* of our original y -coordinate data. Thus, it is the equation $\ln y = r + sx$ that is the least-squares fit of the logarithmic data. Using the least-squares method to fit the logarithm of the y -coordinate data produces an exponential function that approximates the smaller y -values in the data better than it does the larger y -values, as illustrated by Table 6.5. It can be shown that the exponential

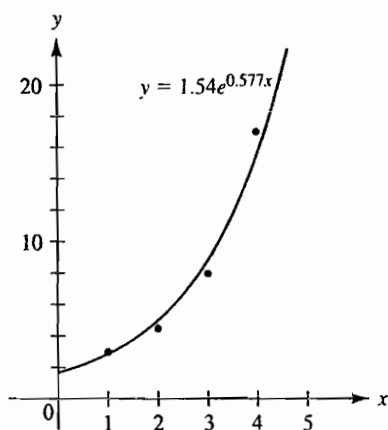


FIGURE 6.16
Data points and the exponential fit.

fit $y = f(x) = re^{sx}$ obtained from the least-squares fit of $\ln y = \ln r + sx$ amounts roughly to minimizing the *percent of error* in taking $f(a_i)$ for b_i . The least-squares fit of logarithmic data does not yield the least-squares fit of the original data. The routine YOUNFIT in LINTEK can be used to illustrate this. (See Exercises 29 and 30.)

Least-Squares Solutions for Larger Systems

Data points of more than two components may lead to overdetermined linear systems in more than two unknowns. Suppose that an experiment is repeated m times, and data values

$$b_i, a_{i1}, \dots, a_{in}$$

are obtained from measurements on the i th experiment. For example, the data values a_{ij} might be ones that can be controlled and the value b_i measured. Of course, there may be errors in the controlled as well as in the measured values. Suppose that we have reason to believe that the data obtained for each experiment should *theoretically* satisfy the same linear relation

$$y = r_0 + r_1x_1 + r_2x_2 + \dots + r_nx_n, \quad (4)$$

with $y = b_i$ and $x_j = a_{ij}$. We then obtain a system of m linear approximations in $n + 1$ unknowns $r_0, r_1, r_2, \dots, r_n$:

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \approx \begin{bmatrix} 1 & a_{11} & a_{12} & \cdots & a_{1n} \\ 1 & a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}. \quad (5)$$

$\mathbf{b} \qquad \qquad \mathbf{A} \qquad \qquad \mathbf{r}$

If $m > n + 1$, then system (5) corresponds to an overdetermined linear system $\mathbf{b} = A\mathbf{r}$, which probably has no exact solution. If the rank of A is $n + 1$, then a repetition of our geometric argument above indicates that the least-squares solution $\bar{\mathbf{r}}$ for the system $\mathbf{b} \approx A\mathbf{r}$ is given by

$$\bar{\mathbf{r}} = (A^T A)^{-1} A^T \mathbf{b}. \quad (6)$$

Again, it is more efficient computationally to solve $(A^T A)\mathbf{r} = A^T \mathbf{b}$.

A linear system of the form shown in system (5) arises if m data points (a_i, b_i) are found and a least-squares fit is sought for a polynomial function

$$y = r_0 + r_1 x + \cdots + r_{n-1} x^{n-1} + r_n x^n.$$

The data point (a_i, b_i) leads to the approximation

$$b_i \approx r_0 + r_1 a_i + \cdots + r_{n-1} a_i^{n-1} + r_n a_i^n.$$

The m data points thus give rise to a linear system of the form of system (5), where the matrix A is given by

$$A = \begin{bmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^n \\ 1 & a_2 & a_2^2 & \cdots & a_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_m & a_m^2 & \cdots & a_m^n \end{bmatrix}. \quad (7)$$

EXAMPLE 3 Use a computer to find the least-squares fit to the data in Problem 2 by a parabola—that is, by a quadratic function

$$y = r_0 + r_1 x + r_2 x^2.$$

SOLUTION We write the data in the form $\mathbf{b} \approx A\mathbf{r}$, where A has the form of matrix (7):

$$\begin{bmatrix} 1 \\ 3 \\ 5 \\ 12 \end{bmatrix} \approx \begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 8 & 64 \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix}.$$

Entering A and $\mathbf{b}$ in either LINTEK or MATLAB, we find that

$$A^T A = \begin{bmatrix} 4 & 19 & 109 \\ 19 & 109 & 709 \\ 109 & 709 & 4993 \end{bmatrix} \quad \text{and} \quad A^T \mathbf{b} = \begin{bmatrix} 21 \\ 135 \\ 945 \end{bmatrix}.$$

Solving the linear system $(A^T A)\mathbf{r} = A^T \mathbf{b}$ using either package then yields

$$\bar{\mathbf{r}} \approx \begin{bmatrix} .207 \\ .010 \\ .183 \end{bmatrix}.$$

Thus, the quadratic function that best approximates the data in the least-squares sense is

$$y \approx f(x) = .207 + .01x + .183x^2.$$

In Figure 6.17, we show the graph of this quadratic function and plot the data points. Data are given in Table 6.6. On the basis of the least-squares fit, we might project that the price of a 10-ton sailing yacht would be about 18.6 times \$10,000, or \$186,000, and that the price of a 20-ton yacht would be about $f(20)$ times \$10,000, or about \$736,000. However, one should be wary of using a function $f(x)$ that seems to fit data well for measured values of x to project data for values of x far from those measured values. Our quadratic function seems to fit our data quite well, but we should not expect the cost we might project for a 100-ton ship to be at all accurate. ■

TABLE 6.6

a_i	b_i	$f(a_i)$
2	1	.959
4	3	3.17
5	5	4.83
8	12	12.0

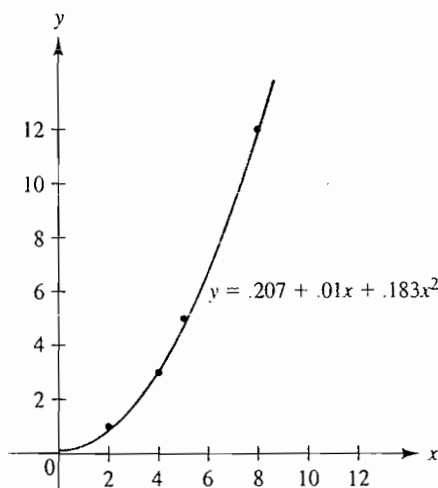


FIGURE 6.17
The graph and data points for Example 3.

EXAMPLE 4 Show that the least-squares linear fit to data points (a_i, b_i) for $i = 1, 2, \dots, m$ by a constant function $f(x) = r_0$ is achieved when r_0 is the average y -value

$$(b_1 + b_2 + \cdots + b_m)/m$$

of the data values b_i .

SOLUTION Such a constant function $y = f(x) = r_0$ has a horizontal line as its graph. In this situation, we have $n = 1$ as the number of unknowns and system (5) becomes

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \approx \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} [r_0].$$

$\mathbf{b} \quad A \quad \mathbf{r}$

Thus A is the $m \times 1$ matrix with all entries 1. We easily find that $A^T A = [m]$, so $(A^T A)^{-1} = [1/m]$. We also find that $A^T \mathbf{b} = b_1 + b_2 + \cdots + b_m$. Thus, the least-squares solution [Eq. (6)] is

$$\bar{r} = (b_1 + b_2 + \cdots + b_m)/m,$$

as asserted. ■

超超超超超

Overdetermined Systems of Linear Equations

We have presented examples in which least-squares solutions were found to systems of linear approximations arising from applications. We used $\mathbf{r}$ as the column vector of unknowns in those examples, because we were using x as the independent variable in the formula for a fitting function $f(x)$. We now discuss the mathematics of overdetermined systems, and we use $\mathbf{x}$ as the column vector of unknowns again.

Consider a general overdetermined system of linear equations

$$A\mathbf{x} = \mathbf{b},$$

where A is an $m \times n$ matrix of rank n and $m > n$. We expect such a system to be inconsistent—that is, to have no solution. As we have seen, the system of linear approximations $A\mathbf{x} \approx \mathbf{b}$ has least-squares solution

$$\bar{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}. \quad (8)$$

The vector $\bar{\mathbf{x}}$ is thus the **least-squares solution of the overdetermined system** $A\mathbf{x} = \mathbf{b}$. Equation (8) can be written in the equivalent form

$$A^T A \bar{\mathbf{x}} = A^T \mathbf{b}, \quad (9)$$

which exhibits $\bar{\mathbf{x}}$ as the solution of the consistent linear system $A^T A \mathbf{x} = A^T \mathbf{b}$.

EXAMPLE 5 Find the least-squares approximate solution of the overdetermined linear system

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 0 \\ 0 & 2 & 1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

by transforming it into the consistent form [system (9)].

SOLUTION The consistent system [system (9)] is

$$\begin{bmatrix} 1 & 2 & 1 & 0 & -1 \\ 0 & 1 & 3 & 2 & 2 \\ 1 & 1 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 0 \\ 0 & 2 & 1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 0 & -1 \\ 0 & 1 & 3 & 2 & 2 \\ 1 & 1 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 7 & 3 & 4 \\ 3 & 18 & 1 \\ 4 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix},$$

whose solution is found, using MATLAB or LINTEK, to be

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \approx \begin{bmatrix} -.614 \\ .421 \\ 1.259 \end{bmatrix},$$

accurate to three decimal places. This is the least-squares approximate solution $\bar{x}$ to the given overdetermined system. ■

The Orthogonal Case

To obtain the least-squares linear fit $y = r_0 + r_1x$ of k data points (a_i, b_i) , we form the matrix

$$A = \begin{bmatrix} 1 & a_1 \\ 1 & a_2 \\ \vdots & \vdots \\ 1 & a_k \end{bmatrix} \quad (10)$$

and compute the least-squares solution vector $\bar{r} = (A^T A)^{-1} A^T \mathbf{b}$. If the set of column vectors in A were orthonormal, then $A^T A$ would be the 2×2 identity matrix I . Of course, this is never the case, because the first column vector of A

has length $\sqrt{k}$. However, it may happen that the column vectors in A are mutually perpendicular; we can see then that

$$A^T A = \begin{bmatrix} k & 0 \\ 0 & \mathbf{a} \cdot \mathbf{a} \end{bmatrix}, \quad \text{so} \quad (A^T A)^{-1} = \begin{bmatrix} 1/k & 0 \\ 0 & 1/(\mathbf{a} \cdot \mathbf{a}) \end{bmatrix}.$$

The matrix A has this property if the x -values $a_1, a_2, \dots, a_k$ for the data are symmetrically positioned about zero. We illustrate with an example.

EXAMPLE 6 Find the least-squares linear fit of the data points $(-3, 8)$, $(-1, 5)$, $(1, 3)$, and $(3, 0)$.

SOLUTION The matrix A is given by

$$A = \begin{bmatrix} 1 & -3 \\ 1 & -1 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

We can see why the symmetry of the x -values about zero causes the column vectors of this matrix to be orthogonal. We find that

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 1 & -1 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 20 \end{bmatrix}.$$

Then

$$\begin{aligned} \bar{\mathbf{r}} = \begin{bmatrix} r_0 \\ r_1 \end{bmatrix} &= (A^T A)^{-1} A^T \mathbf{b} = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{20} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -3 & -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \\ 3 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{20} \end{bmatrix} \begin{bmatrix} 16 \\ -26 \end{bmatrix} = \begin{bmatrix} 4 \\ -1.3 \end{bmatrix}. \end{aligned}$$

Thus, the least-squares linear fit is given by $y = 4 - 1.3x$. ■

Suppose now that the x -values in the data are not symmetrically positioned about zero, but that there is a number $x = c$ about which they are symmetrically located. If we make the variable transformation $t = x - c$, then the t -values for the data are symmetrically positioned about $t = 0$. We can find the t, y -equation for the least-squares linear fit, and then replace t by $x - c$ to obtain the x, y -equation. (Exercises 14 and 15 refine this idea still further.)

EXAMPLE 7 The number of sales of a particular item by a manufacturer in each of the first four months of 1995 is given in Table 6.7. Find the least-squares linear fit of these data, and use it to project sales in the fifth month.

TABLE 6.7

a_i = Month	1	2	3	4
b_i = Thousands sold	2.5	3	3.8	4.5

SOLUTION Our x -values a_i are symmetrically located about $c = 2.5$. If we take $t = x - 2.5$, the data points (t, y) become

$$(-1.5, 2.5), \quad (-.5, 3), \quad (.5, 3.8), \quad (1.5, 4.5).$$

We find that

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1.5 & -.5 & .5 & 1.5 \end{bmatrix} \begin{bmatrix} 1 & -1.5 \\ 1 & -.5 \\ 1 & .5 \\ 1 & 1.5 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}.$$

Thus,

$$\begin{aligned} (A^T A)^{-1} A^T \mathbf{b} &= \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1.5 & -.5 & .5 & 1.5 \end{bmatrix} \begin{bmatrix} 2.5 \\ 3 \\ 3.8 \\ 4.5 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 13.8 \\ 3.4 \end{bmatrix} = \begin{bmatrix} 3.45 \\ .68 \end{bmatrix}. \end{aligned}$$

Thus, the t, y -equation is $y = 3.45 + .68t$. Replacing t by $x - 2.5$, we obtain

$$y = 3.45 + .68(x - 2.5) = 1.75 + .68x.$$

Setting $x = 5$, we estimate sales in the fifth month to be 5.15 units, or 5150 items. ■

Using the QR -Factorization

The QR -factorization discussed at the end of Section 6.2 has application to the method of least squares. Consider an overdetermined system $A\mathbf{x} = \mathbf{b}$, where the matrix A has independent column vectors. We factor A as QR and remember that $Q^T Q = I$, because Q has orthonormal column vectors. Then we obtain

$$\begin{aligned} (A^T A)^{-1} A^T \mathbf{b} &= ((QR)^T QR)^{-1} (QR)^T \mathbf{b} = (R^T Q^T QR)^{-1} R^T Q^T \mathbf{b} \\ &= (R^T R)^{-1} R^T Q^T \mathbf{b} = R^{-1} (R^T)^{-1} R^T Q^T \mathbf{b} = R^{-1} Q^T \mathbf{b}. \end{aligned} \quad (11)$$

To find the least-squares solution vector $R^{-1} Q^T \mathbf{b}$ once Q and R have been found, we first multiply $\mathbf{b}$ by Q^T , which is a stable computation because Q is orthogonal. We then solve the upper-triangular system

$$R\mathbf{x} = Q^T \mathbf{b} \quad (12)$$

by back substitution. Because R is already upper triangular, no matrix reduction is needed! The routine QRFACTOR in LINTEK has an option for this least-squares computation. The program demonstrates how quickly the least-squares solution vector can be found once a QR -factorization of A is known.

Suppose we believe that the output y of a process should be a linear function of time t . We measure the y -values at specified times during the process. We can then find the least-squares linear fit $y = r_1 + s_1 t$ from the measured data by finding the QR -factorization of an appropriate matrix A and subsequently finding the solution vector [Eq. (11)] in the way we just described. If we repeat the process later and make measurements again at the same time intervals, the matrix A will not change, so we can find another least-squares linear fit using the solution vector with the same matrices Q and R . We can repeat this often, obtaining linear fits $r_1 + s_1 t, r_2 + s_2 t, \dots, r_m + s_m t$. For our final estimate of the proper linear fit for this process, we might take $r + st$, where r is the average of the m values r_i and s is the average of the n values s_i . We expect that errors made in measurements will roughly cancel each other out over numerous repetitions of the process, so that our estimated linear fit $r + st$ will be fairly accurate.

SUMMARY

1. A set of k data points (a_i, b_i) whose first coordinates are not all equal can be fitted with a polynomial function or with an exponential function, using the method of least squares illustrated in Examples 1 through 3.
2. Suppose that the x -values a_i in a set of k data points (a_i, b_i) are symmetrically positioned about zero. Let A be the $k \times 2$ matrix with first column vector having all entries 1 and second column vector $\mathbf{a}$. The columns of A are orthogonal, and

$$A^T A = \begin{bmatrix} k & 0 \\ 0 & \mathbf{a} \cdot \mathbf{a} \end{bmatrix}.$$

Computation of the least-squares linear fit for the data points is then simplified. If the x -values of the data points are symmetrically positioned about $x = c$, then the substitution $t = x - c$ gives data points with t -values symmetrically positioned about zero, and the above simplification applies. See Example 7.

Let $A\mathbf{x} = \mathbf{b}$ be a linear system of m equations in n unknowns, where $m > n$ (an overdetermined system) and where the rank of A is n .

3. The least-squares solution of the corresponding system $A\mathbf{x} \approx \mathbf{b}$ of linear approximations is the vector $\mathbf{x} = \bar{\mathbf{x}}$, which minimizes the magnitude of the error vectors $A\mathbf{x} - \mathbf{b}$ for $\mathbf{x} \in \mathbb{R}^n$.

4. The least-squares solution $\bar{x}$ of $Ax \approx b$ and of $Ax = b$ is given by the formula

$$\bar{x} = (A^T A)^{-1} A^T b.$$

Geometrically, $A\bar{x}$ is the projection of b on the column space of A .

5. An alternative to using the formula for $\bar{x}$ in summary item 4 is to convert the overdetermined system $Ax = b$ to the consistent system $(A^T A)x = A^T b$ by multiplying both sides of $Ax = b$ by A^T , and then to find its unique solution, which is the least-squares solution $\bar{x}$ of $Ax = b$.

EXERCISES

- Let the length b_i of a spring with an attached weight a_i be determined by measurements, as shown in Table 6.8.
 - Find the least-squares linear fit, in accordance with Hooke's law.
 - Use the answer to part (a) to estimate the length of the spring if a weight of 5 ounces is attached.

TABLE 6.8

a_i = Weight in ounces	1	2	4	6
b_i = Length in inches	3	4.1	5.9	8.2

- A company had profits (in units of \$10,000) of 0.5 in 1989, 1 in 1991, and 2 in 1994. Let time t be measured in years, with $t = 0$ in 1989.
 - Find the least-squares linear fit of the data.
 - Using the answer to part (a), estimate the profit in 1995.
- Repeat Exercise 2, but find an exponential fit of the data, working with logarithms as explained in Example 2.
- A publishing company specializing in college texts starts with a field sales force of ten people, and it has profits of \$100,000. On increasing this sales force to 20, it has profits of \$300,000; and increasing its sales force to 30 produces profits of \$400,000.
 - Find the least-squares linear fit for these data. [Hint: Express the numbers of salespeople in multiples of 10 and the profit in multiples of \$100,000.]

- Use the answer to part (a) to estimate the profit if the sales force is reduced to 25.
- Does the profit obtained using the answer to part (a) for a sales force of 0 people seem in any way plausible?

In Exercises 5–7, find the least-squares fit to the given data by a linear function $f(x) = r_0 + r_1 x$. Graph the linear function and the data points.

- (0, 1), (2, 6), (3, 11), (4, 12)
- (1, 1), (2, 4), (3, 6), (4, 9)
- (0, 0), (1, 1), (2, 3), (3, 8)
- Find the least-squares fit to the data in Exercise 6 by a parabola (a quadratic polynomial function).
- Repeat Exercise 8, but use the data in Exercise 7 instead.

In Exercises 10–13, use the technique illustrated in Examples 6 and 7 to solve the least-squares problem.

- Find the least-squares linear fit to the data points $(-4, -2)$, $(-2, 0)$, $(0, 1)$, $(2, 4)$, $(4, 5)$.
- Find the least-squares linear fit to the data points $(0, 1)$, $(1, 4)$, $(2, 6)$, $(3, 8)$, $(4, 9)$.
- The gallons of maple syrup made from the sugar bush of a Vermont farmer over the past five years were:
80 gallons five years ago,
70 gallons four years ago,
75 gallons three years ago,
65 gallons two years ago,
60 gallons last year.

